Geometry Review: Unit 6 — Circles

Station 1 of 4: Parts of a Circle, Circumference, and Area

Name: _____ Date: _____

Leave answers in terms of π unless told to use an approximation. Diagrams are not drawn to scale.

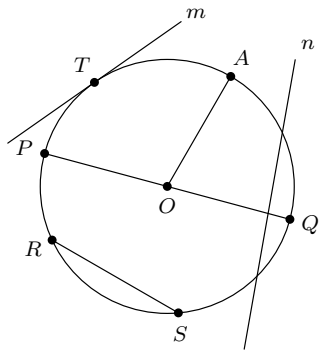
1. A circle has radius 5. Find its circumference and area.

2. A circle has diameter 14. Find its radius, circumference, and area.

3. A circle has circumference 18π. Find its radius and area.

4. A circle has area 64π. Find its radius and diameter.

5. Use the circle with center O. Classify each as a radius, diameter, chord, tangent, or secant: (a) $\overline{OA}$ (b) $\overline{PQ}$ (c) $\overline{RS}$ (d) line m (e) line n.



6. A circular garden has a diameter of 20 ft. Using $\pi \approx 3.14$, how much fencing is needed to go around it?

7. A 12-inch pizza and a 16-inch pizza (diameters). How much more area does the larger pizza have?

8. If the radius of a circle is doubled, what happens to its area?

9. A wheel has radius 14 in. How far does it roll in one full turn? Give the exact answer and a value to the nearest inch using $\pi \approx 3.14$.

10. A ring is bounded by circles of radii 9 and 5. Find the area of the ring.

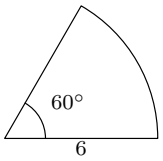
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Station 2 of 4: Arc Length and Sector Area

Name: _____ Date: _____

Use arc length $= \frac{\theta}{360} \cdot 2\pi r$ *and sector area* $= \frac{\theta}{360} \cdot \pi r^2$*. Leave answers in terms of* π*. Diagrams are not drawn to scale.*

1. $r = 6$, central angle $60°$. Find the arc length and sector area.



2. $r = 10$, central angle $90°$. Find the arc length and sector area.

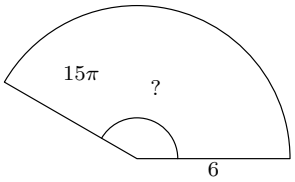
3. $r = 8$, central angle $45°$. Find the arc length and sector area.

4. $r = 9$, central angle $120°$. Find the arc length and sector area.

5. $r = 12$, central angle $150°$. Find the arc length and sector area.

6. An arc of length 4π is in a circle of radius 6. Find the central angle.

7. A sector of area 15π is in a circle of radius 6. Find the central angle.



8. $r = 5$, central angle $72°$. Find the arc length and sector area.

9. A pizza has diameter 16 in and is cut into 8 equal slices. Find the area of one slice.

10. An arc with a central angle of $90°$ has length 6π. Find the radius of the circle.

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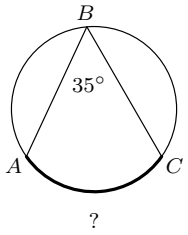
Station 3 of 4: Central Angles, Inscribed Angles, and Arcs

Name: _____ Date: _____

Use arc and angle relationships. Include degree symbols. Diagrams are not drawn to scale.

1. A central angle measures 80°. What is the measure of its intercepted arc?

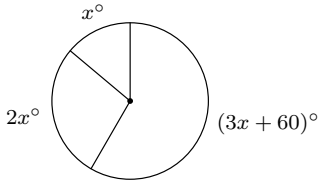
2. An inscribed angle measures 35°. What is the measure of its intercepted arc?



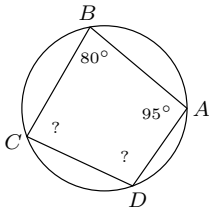
3. An arc measures 124°. What is the measure of an inscribed angle that intercepts it?

4. An inscribed angle intercepts a diameter (a semicircle). What is its measure?

5. Three arcs that make up a circle measure $x°$, $2x°$, and $(3x + 60)°$. Find each arc.

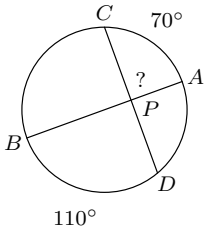


6. A quadrilateral is inscribed in a circle. Two consecutive angles measure 95° and 80°. Find the other two angles.

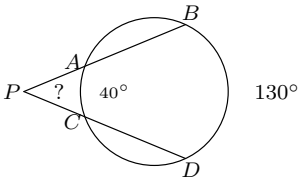


7. Two inscribed angles intercept the same arc. One measures 41°. Find the other.

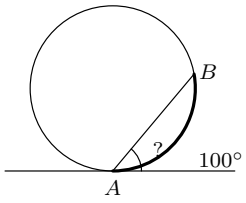
8. Two chords intersect inside a circle and intercept arcs of 70° and 110°. Find the angle formed.



9. Two secants meet outside a circle and intercept arcs of 130° and 40°. Find the angle formed.



10. A tangent and a chord meet at a point on the circle and intercept an arc of 100°. Find the angle formed.



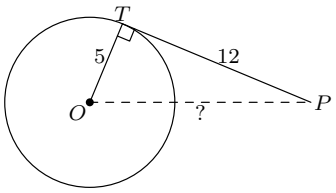
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Station 4 of 4: Chords, Tangents, and Segment Lengths

Name: _____ Date: _____

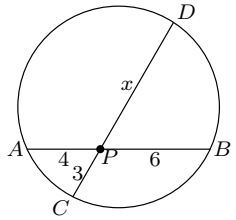
Use circle theorems about chords, tangents, and secants. Show your equations. Diagrams are not drawn to scale.

1. A tangent segment from an external point is 12 long and the radius is 5. Find the distance from the external point to the center.



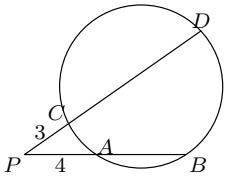
2. Two tangent segments from point P to a circle are $PA = 3x + 4$ and $PB = 5x - 6$. Find x and PA.

3. Two chords intersect inside a circle. The segments of one chord are 4 and 6; the segments of the other are 3 and x. Find x.

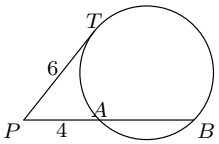


4. Two chords intersect inside a circle. The segments of one chord are 2 and x; the segments of the other are 5 and 6. Find x.

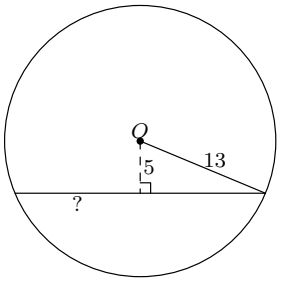
5. Two secants are drawn from an external point. The first has external part 4 and whole length 9. The second has external part 3 and whole length x. Find x.



6. A tangent of length 6 and a secant are drawn from the same external point. The secant's external part is 4. Find the whole secant and its part inside the circle.



7. The radius of a circle is 13. A chord is 5 units from the center. Find the length of the chord.



8. A chord of length 16 is in a circle of radius 10. How far is the chord from the center?

9. Chords $\overline{AB}$ and $\overline{CD}$ are the same distance from the center. $AB = 3x - 4$ and $CD = 2x + 5$. Find x and AB.

10. A tangent segment from a point 17 units from the center of a circle has length 15. Find the radius.

Geometry Review: Unit 6 — Circles

Answer Key

Station 1: Parts of a Circle, Circumference, and Area

1. $C = 10\pi,\ A = 25\pi$
2. $r = 7,\ C = 14\pi,\ A = 49\pi$
3. $r = 9,\ A = 81\pi$
4. $r = 8,\ d = 16$
5. (a) radius; (b) diameter; (c) chord; (d) tangent (touches the circle at one point); (e) secant (crosses the circle twice)
6. $C = 20\pi \approx 62.8$ ft
7. $64\pi - 36\pi = 28\pi$ in^2
8. The area is multiplied by 4
9. $28\pi \approx 88$ in
10. $81\pi - 25\pi = 56\pi$

Station 2: Arc Length and Sector Area

1. $2\pi;\ 6\pi$
2. $5\pi;\ 25\pi$
3. $2\pi;\ 8\pi$
4. $6\pi;\ 27\pi$
5. $10\pi;\ 60\pi$
6. $120°$
7. $150°$
8. $2\pi;\ 5\pi$
9. 8π in^2
10. 12

Station 3: Central Angles, Inscribed Angles, and Arcs

1. $80°$
2. $70°$
3. $62°$
4. $90°$
5. $50°,\ 100°,\ 210°$
6. $85°$ and $100°$ *opposite angles are supplementary*
7. $41°$
8. $90°$ $\quad \frac{70 + 110}{2}$
9. $45°$ $\quad \frac{130 - 40}{2}$
10. $50°$

Station 4: Chords, Tangents, and Segment Lengths

1. 13 $\quad 5^2 + 12^2 = 13^2$
2. $x = 5;\ PA = 19$
3. $x = 8 \quad 4 \cdot 6 = 3x$
4. $x = 15$
5. $x = 12 \quad 4 \cdot 9 = 3x$
6. 9; 5 $\quad 6^2 = 4 \cdot w$
7. 24 $\quad$ *half-chord* $= \sqrt{13^2 - 5^2} = 12$
8. 6
9. $x = 9;\ AB = 23$
10. 8